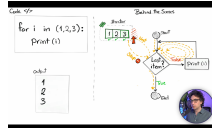
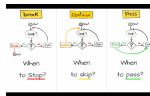


A for loop in Python is a controlled flow statement used for iterating over a sequence (such as a list, tuple, string or range) or other iterable objects. It executes a block of code once for each item in the sequence, making it ideal when you know the number of iterations in advance.

Basic Syntax:
 for variable in sequence:
 # Code block to execute repeatedly
 # This block must be indented!



optional
 default = 0
 for
 range (start, stop, step)
 default = 1

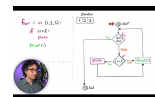


break statement:

The break statement in Python is a loop control statement used to immediately terminate the execution of the current loop (either a for or while loop) and transfer program control to the next line of code following the loop.

pass statement:

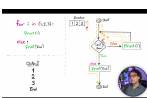
The pass statement in Python is a null operation; nothing happens when it executes. It is used as a placeholder where a statement is syntactically required by Python's grammar, but you do not want any code to run yet.



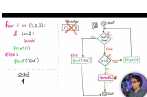
BREAK VS CONTINUE VS PASS



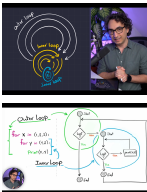
for else:



break + else:



Nested loops:



while loop:

A while loop in Python repeatedly executes a block of code as long as a specified condition remains True. This loop type is ideal when the number of iterations is not known in advance, such as when processing user input or waiting for a specific event.

